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The two-stage machine learning ensemble models for stock price prediction by combining mode decomposition, extreme learning machine and improved harmony search algorithm

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(VMD)), extreme learning machine (ELM) and improved harmony search (IHS) algorithm for stock price prediction, which are respectively named EMD–ELM–IHS and VMD–ELM–IHS. Furthermore, to demonstrate the efficiency and performance of the proposed models, the results were compared with those obtained by other methods, including EMD based ELM (EMD–ELM), VMD based ELM (VMD–ELM), autoregressive integrated moving average (ARIMA), ELM, multi-layer perception (MLP), support vector regression (SVR), and long short-term memory (LSTM) models. The results show that the proposed models have superior performance in terms of its accuracy and stability as compared to the other models. Also, we find that the sizes of sliding window and training set have a significant impact on the predictive performance.

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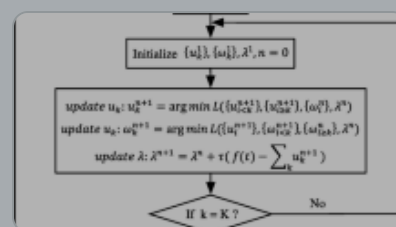
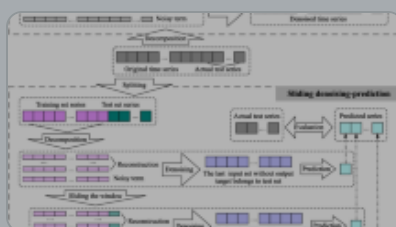
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